

Starting Stopping

Clay Smalley

The horn is unique among instruments in that the hand is placed in the bell while playing. Historically, this was done on valveless, or *natural*, horns, to reach pitches in between those along the harmonic series. Nowadays, on valve horns, the right hand is mostly used to make intonation and timbre adjustments, as well as playing stopped and echo tones.

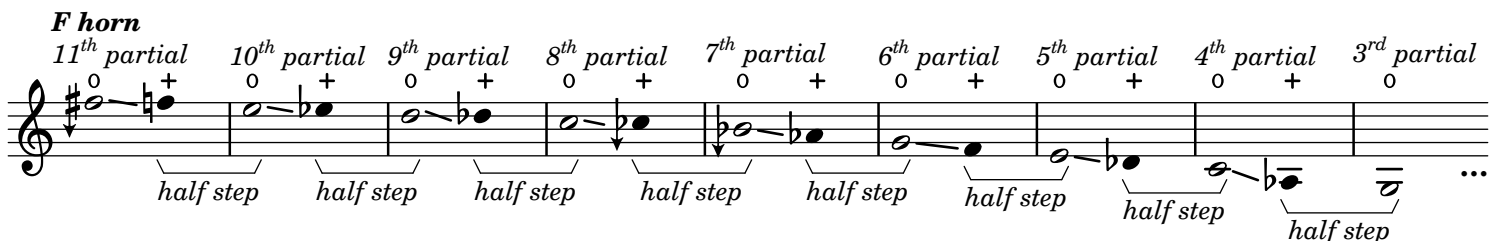
Stopped tones are notated as $\overset{+}{\text{♩}}$, *stopped*, *gestopft* (German), *bouché* (French) or *chiuso* (Italian). A return to normal playing technique is indicated with $\overset{0}{\text{♩}}$, *open*, *offen* (German), *ouvert* (French) or *aperto* (Italian). Echo tones are less common, and accordingly, notation for them is not as well established. In this work, $\overset{\oplus}{\text{♩}}$ is used to indicate echo tones.

An everlasting source of confusion for new hornists is whether stopping the bell *raises* or *lowers* the pitch produced. By playing a long tone and gradually closing the bell, one may notice that the pitch bends down, and one can find a “mostly stopped” position that reliably flattens the horn by a half step. But by sealing off as much of the bell as possible and playing with more pressure, one may notice that the harmonic series has shifted a half step *upward*:

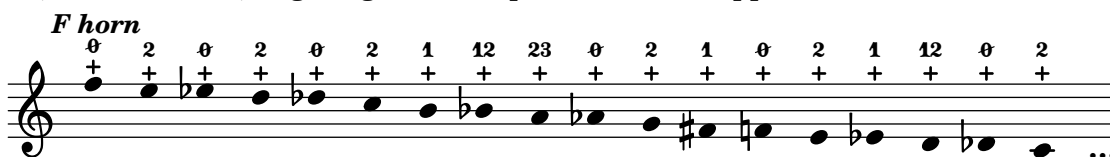


So what is happening here? Each open tone on the F horn indeed has a corresponding stopped tone a half step up, but an open pitch can't be bent upward. The stopped tone actually comes from bending down the next highest partial—a barber pole effect. Closing the bell lowers the pitch, all the way down to *one half step above the next lowest partial* (or, on the B $\flat$ horn, a noticeably out-of-tune $\frac{3}{4}$ step above it).

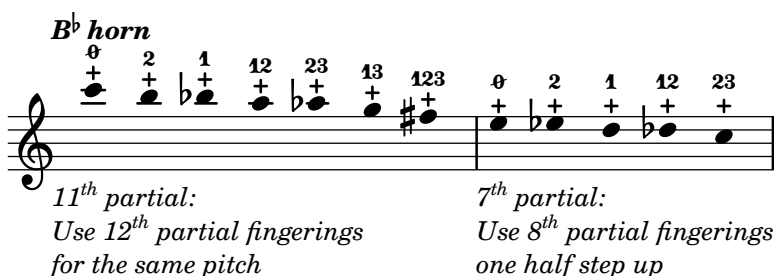
For example, the 5th and 6th partials produce an open E and G, respectively. One half step above the 5th partial's E is F, which is the stopped tone that the 6th partial's G can be bent down to:



So, on the F horn, fingering a half step down while stopped is a useful mnemonic:



Though a limited set of fingerings can counteract the out-of-tune 7th and 11th partials and bring the stopped B $\flat$ horn back into tune. Since these partials are rarely used otherwise, mnemonics based on more recognizable fingerings for the 8th and 12th partials, respectively, are more common:



A Short Pitch Bends

Begin by playing the pattern on valve horn, using the suggested fingerings with the bell open ($\emptyset$). Upon returning to the upper pitch, gradually transition from open to stopped ($\overset{+}{\emptyset}$) and back, applying more pressure when stopped. Match intonation between open and stopped tones.

$\text{♩} = 120 - 176$

B $\flat$ horn

0 2 $\emptyset$ + 0 2 1 2 + 0

F horn

1 12 1 + 0 12 23 12 + 0 23 $\emptyset$ 23 + 0

7

F horn

1 2 1 + 0 12 1 12 + 0 23 12 23 + 0

16

The out-of-tune 7th partial is used here to reach certain stopped tones.

$\emptyset$ 23 $\emptyset$ + 0 2 $\emptyset$ 2 + 0 1 2 1 + 0

25

B Long Pitch Bends

As above, transitioning from open to echo ($\emptyset$) to stopped and back. Match intonation between all tones.

$\text{♩} = 120 - 176$

F horn

0 2 1 2 $\emptyset$ + $\emptyset$ 0 2 1 $\emptyset$ 1 2 + $\emptyset$ 0

F horn

1 $\emptyset$ 2 $\emptyset$ 1 + $\emptyset$ 0 12 2 1 2 12 + $\emptyset$ 0

11

F horn

$\emptyset$ 2 12 2 $\emptyset$ + $\emptyset$ 0 2 1 $\emptyset$ 1 2 + $\emptyset$ 0

21

C Alternating Open and Stopped

Match intonation between open ($\overset{0}{\text{O}}$) and stopped ($\overset{+}{\text{O}}$) tones.

$\text{♩} = 76 - 120$

f

9

17

25

33

41

49

57

65